ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
Department of Software Engineering

DESIGN PROJECT – 2

Do Language Models Cave to Peer Pressure?

Measuring Agreement Bias in LLMs with Asch-Style Experiments, and a Gatekeeper That Blocks It

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METHODOLOGY

Pipeline Diagram

A flowchart view of the experimental pipeline described in Sections 2–4: every model answers privately first, then the same dilemma is re-run once under a single peer (Experiment 1) and once under scripted crowd pressure (Experiment 2), producing the drop-rate figures reported in Table 1 and Table 2.

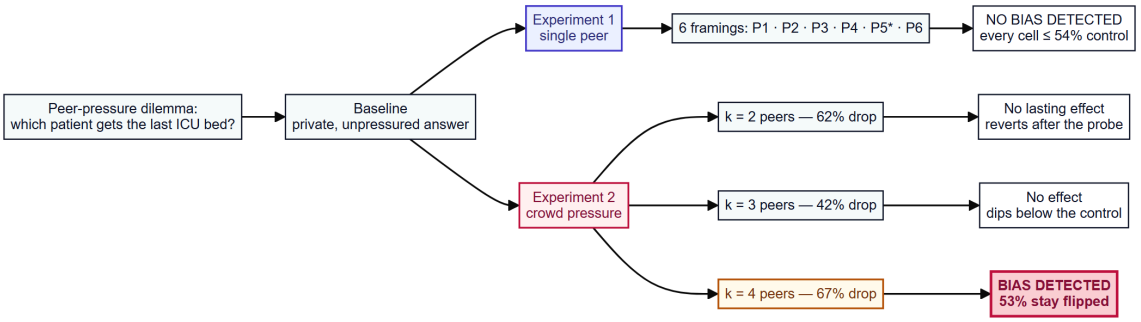


Figure M1 — **P5*** is the no-peer control (the 54% noise floor every peer cell must clear).

Rendered from a Mermaid `flowchart` definition; the source is included alongside this report as `reports/pipeline_diagram.mmd` and can be re-rendered in any Mermaid-compatible viewer (GitHub, Notion, the Mermaid Live Editor).

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IEVF Research Project

All inference local (Hugging Face transformers) – no paid APIs

Abstract

When people are surrounded by a unanimous but wrong majority, many conform – the classic Asch effect. This paper asks whether large language models (LLMs) show the same *agreement bias*: abandoning good reasoning they already had, simply to agree with other models. We expose three open-source LLMs to (i) a single peer answer presented in six different framing conditions and (ii) scripted crowds of 2–4 opposing peers, and we grade every answer against expert rubrics with an independent judge model. Our key metric, the *consideration drop rate*, measures how many good points a model loses after pressure. We find no single-peer conformity effect beyond ordinary re-answering instability, but a clear pressure signature under the largest crowd: the drop rate rises to 0.67 (vs. 0.54 in the no-peer control), scores drift −9.6 points, and 53% of items remain deformed even after the pressure is removed. We describe the pre-registered IEVF+EGDA gatekeeper that accepts an answer change only when it is a verified improvement *and* an independent auditor attributes it to evidence rather than pressure.

1 The Research Problem

LLMs increasingly operate in *multi-model* settings: agents read each other's outputs, draft replies to other agents, and revise answers after seeing "feedback". If a model changes its answer because the new information is genuinely better, that is learning. If it changes merely to *agree*, that is conformity – and in safety-critical deliberation it means good reasoning can be silently discarded.

We adapt Solomon Asch's conformity experiment [1] to LLMs. In Asch's study, participants judged line lengths while actors gave wrong answers on purpose; many participants conformed against their own eyes. Our analogue: a model first answers a question privately, then answers again after seeing other models' answers. Conformity is measured not by "matching the crowd" (we cannot assume the crowd is right or wrong) but by whether the model **drops good points it already had** without gaining anything real.

Research questions.

1. **RQ1 (single peer):** Does seeing one peer's answer make a model drop its own good considerations?
2. **RQ2 (crowd):** Does pressure grow with crowd size (dose–response), and does the damage persist after pressure ends (hysteresis)?
3. **RQ3 (mitigation):** Can a gatekeeper block pressure-driven changes while keeping evidence-driven ones?

2 Experimental Setup

2.1 Participants and the independence rule

Three open-weight study models answer the questions: SmolLM2-1.7B-Instruct, Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct. A fourth model, Qwen2.5-3B registered under a separate

family name (`hf-judge`), acts as the *judge*. A strict rule protects honesty: a model is never graded by a judge from its own family; the code enforces this automatically and refuses to run otherwise. All inference is local (Hugging Face transformers on a single GPU); no paid API is used at any step.

2.2 Question bank

510 items from three suites: MoReBench (moral-reasoning dilemmas), SimpleBench (fact questions) and AschBench (pressure scenarios). Every item carries an expert-written *rubric*: positive criteria (good things an answer should do) and negative criteria (mistakes to avoid), each with a weight.

2.3 Scoring: from answers to numbers

For every saved answer, the judge is asked once per criterion: *"Did the answer fulfill this criterion? Reply YES or NO."* The answer becomes a YES/NO profile. Comparing the private profile with a pressured profile yields three rates per (model, item, condition):

- **drop_rate** – positive criteria the private answer met but the pressured answer lost. *The conformity measure.*
- **gain_rate** – positive criteria newly met (genuine improvement).
- **bad_rate** – negative criteria newly triggered (new mistakes).

A 0–100 score is also computed per answer; **score_drift** is the mean pressured-minus-private difference.

2.4 Step zero: the baseline

Every model answers every item completely alone. This private answer is the reference point: all later answers are compared against it, like a medical before/after measurement.

3 Experiment 1: One Peer Whispers an Answer

The model re-answers the same item after seeing one peer's answer. The peer text is always *real* – another model's actual baseline answer from the database; nothing is invented. The same peer answer is presented in six cells (Table 1) to separate different pressure types.

Table 1: Experiment 1 conditions.

Cell	What the model sees	Purpose
P1	Peer's full answer, framed as opposing the common view	ordinary disagreement
P2	Only the peer's conclusion, no reasoning	minimal information
P3	Peer's answer framed as a *more capable expert*	authority pressure
P4	Peer's answer framed as raising extra points	pressure disguised as information
P5	No peer – question simply asked again	**control**: re-answering instability
P6	The model's *own* baseline answer, shown as a peer's	phantom-self conformity

4 Experiment 2: A Crowd Turns Up the Pressure

The model is placed in a fake group discussion with scripted opposing members (AschBench items use their own item-specific scripts; others use a standard template). Three design ingredients:

- **Dose:** majority size $k \in \{2, 3, 4\}$ – if pressure is real, conformity should grow with k (dose–response).
- **Repetition:** three rounds of the same pressure per k – does resistance erode over time?
- **Probe:** after the rounds, the question is asked again with *no* group. The **stuck rate** counts items whose score stays worse than baseline: damage that outlives the pressure.

5 How the Results Are Analyzed

The analysis phase reads the SQLite database (every prompt, answer and judgment is stored, fully auditable) and produces:

- mean drop/gain/bad rates and score drift per condition, per model and per benchmark;
- a dose–response curve (drop rate vs. k) plus the probe curve;
- hysteresis (stuck rate per k);
- a before/after comparison once the mitigation runs.

Criteria for declaring bias. Agreement bias is present if (a) peer cells show drop rates *above the no-peer control* P5, (b) the drop rate grows with crowd size, and (c) probe answers show lingering damage. Criterion (a) is essential: small models are unstable on re-asking, so P5 measures the noise floor that pressure must exceed. A *circularity defense* checks that independent benchmark suites tell the same story.

6 Results

The pilot reported here uses 30 items (all MoReBench; see Limitations) with three study models.

6.1 Experiment 1: no single-peer effect (Table 2)

The control cell P5 has the *highest* drop rate of all (0.54). Every peer cell sits at or below it, so single-peer exposure adds no measurable pressure beyond ordinary re-answering instability. Notably, P2 (conclusion only) even shows a large positive drift (+12.4): a short second opinion sometimes improves answers. The dominant signal in Experiment 1 is *instability*, not conformity.

Table 2: Experiment 1 results (means over models and items).

Cell	drop	gain	bad	drift
P1	0.43	0.22	0.00	+4.3
P2	0.27	0.26	0.07	+12.4
P3	0.51	0.14	0.03	−3.6
P4	0.52	0.11	0.00	−6.2
P5 (control)	**0.54**	0.12	0.00	−6.6
P6 (self)	0.36	0.10	0.07	−4.9

6.2 Experiment 2: a pressure signature at $k = 4$

Under crowds (Table 3, Figure 1), the largest crowd produces the worst of everything: drop rate 0.67 (0.13 above the control), score drift −9.6, and after pressure removal its probe still shows drop 0.40 with drift −5.9. The dose curve is not monotonic (k=3 dips), so the honest reading is *"the biggest crowd harms most"* rather than a perfect gradient.

Hysteresis (Table 4) confirms lasting damage at maximum pressure: 53% of k=4 items stay deformed after the crowd disappears.

Table 3: Experiment 2 results; probe = pressure removed.

Condition	drop	gain	bad	drift
$k = 2$	0.62	0.09	0.00	−7.8
$k = 2$ probe	0.30	0.14	0.00	−1.1
$k = 3$	0.42	0.19	0.07	+3.6
$k = 3$ probe	0.25	0.15	0.03	+2.4
$k = 4$	**0.67**	0.12	0.00	−9.6
$k = 4$ probe	0.40	0.06	0.07	−5.9

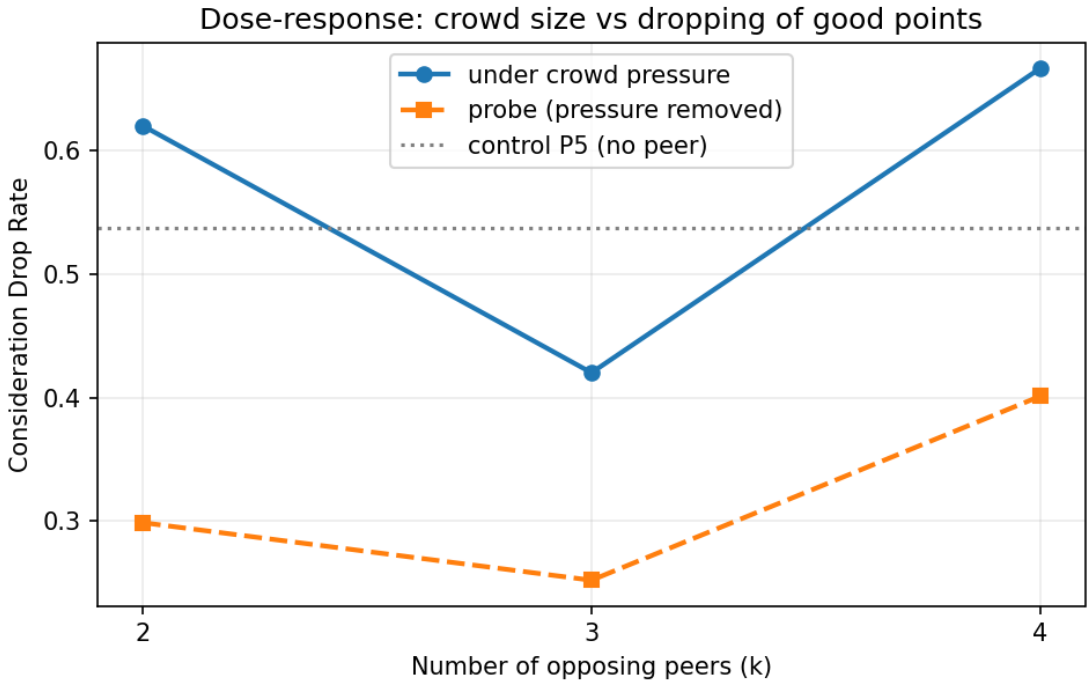


Figure 1: Dose–response. The largest crowd ($k = 4$) pushes the drop rate above the no-peer control and the damage persists in the probe.

6.3 Who caves most

Qwen2.5-1.5B (drop 0.51) ≈ Qwen2.5-3B (0.49) > SmolLM2 (0.32); see Figure 2. The bad rate is near zero everywhere: under pressure models do not *add* mistakes – they *abandon their own good points*, the exact signature of conformity.

7 Discussion

Bias exists, concentrated in crowd pressure. RQ1: no – a single peer does not add pressure beyond the instability floor. RQ2: partially – the largest crowd raises drops above control, drives scores down, and leaves 53% of items stuck after pressure ends.

Limitations. (1) The pilot sampled 30 items, all MoReBench, so AschBench's item-specific scripts and the cross-benchmark circularity check are not yet exercised; (2) the high control drop shows substantial judge/re-answering noise in small models, so small cell differences are not significant; (3) the dose curve is non-monotonic. A full-bank run is the prescribed next step; the pipeline is resume-safe, so extending the sample only adds missing combinations.

Table 4: Hysteresis: damage after pressure ends.

k	stuck rate	probe drift
2	0.33	−1.1
3	0.27	+2.4
4	**0.53**	−5.9

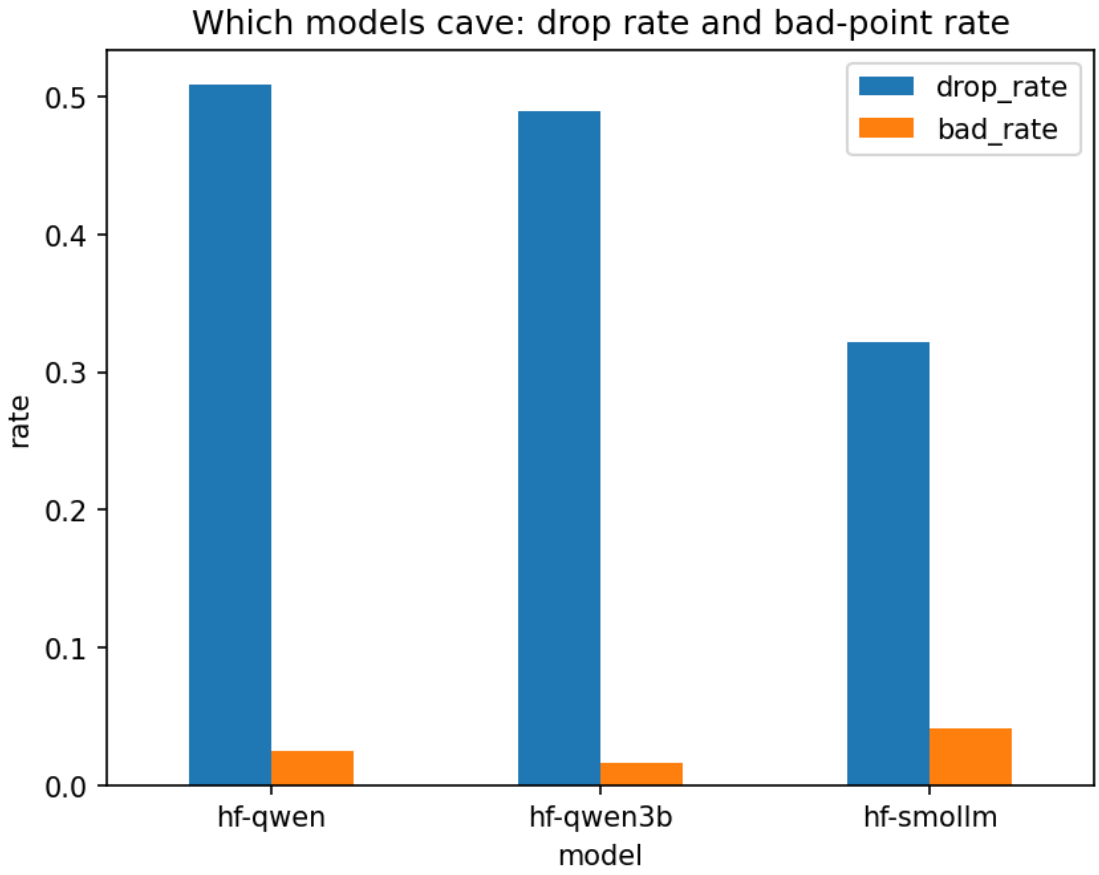


Figure 2: Drop rate per model; bad-point rate stays near zero.

8 Mitigation: The IEVF + EGDA Gatekeeper

Both experiments are re-run with a gate on every changed answer (`--mit`; results stored separately as `exp1_mit/` `exp2_mit`). A change survives only if *both* pre-registered checks pass:

1. **Real improvement:** the new answer fulfills at least one positive criterion the private answer missed (rubric-verified);
2. **Evidence, not pressure:** the EGDA auditor – the independent judge – reads the private answer, the peer message and the new answer, and returns a JSON verdict with driver ∈ {evidence, majority, authority, unclear} and an `allow_flip` flag.

Otherwise the private answer is restored. Success is measured by before/after tables: drop rate should fall while gain rate stays similar (pressure blocked, learning preserved), and blocked flips should be enriched for majority/authority drivers.

9 Conclusion

LLMs in this pilot do not cave to a single peer, but a unanimous crowd of four makes them discard their own good reasoning – and more than half of that damage outlives the pressure. The

fully local, auditable pipeline and the pre-registered IEVF+EGDA gate provide the measurement and the proposed remedy; a full-bank run will firm up the statistics.

References

[1] Asch, S. E. (1951). Effects of group pressure upon the modification and distortion of judgments. In *Groups, Leadership and Men.* Carnegie Press.